EX NO: 1 PROBLEM STATEMENT FOR HOSPITAL MANAGEMENT SYSTEM

Aim:

To write a problem statement for a Hospital Management System.

Problem Statement:

The Hospital Management System (HMS) will interface with various modules and hardware/software components, including:

- Reception module for patient registration and inquiries
- Doctor's console for viewing patient records, prescribing treatments, and scheduling
- Patient console (web or kiosk-based) for appointment booking, billing status, and medical history
- Pharmacy system for issuing prescribed medicines
- Laboratory module for sample collection, testing, and reporting
- Billing system for processing all financial transactions
- Admin dashboard for hospital staff to control the workflow and manage hospital logistics
- Emergency module for handling urgent admissions and critical cases
- The HMS will also maintain secure and real-time communication between departments and synchronize data with the hospital's centralized database.

Objectives

The goal of the software is to streamline and digitize the entire hospital workflow. It ensures the smooth operation of administrative and clinical processes, minimizes paperwork, reduces waiting time for patients, and ensures accurate record-keeping. The system aims to:

- Improve patient care and service efficiency
- Simplify record management and access to patient history
- Automate billing, payments, and pharmacy operations
- Provide detailed reports for hospital management and government compliance

Core Functionalities:

The Hospital Management System must be able to perform the following services:

1. Patient Registration and Management

- Allow staff to register new patients (inpatients or outpatients)
- Assign unique patient IDs
- Record demographics, contact info, insurance details, and emergency contacts

2. Appointment Scheduling

- Enable patients to schedule or reschedule appointments with specific doctors
- Notify doctors of their daily schedule and changes
- Handle walk-in patient queuing efficiently.

3. Doctor Management
    - Maintain doctor profiles and schedules
    - Allow doctors to access patient history and input diagnoses, test requests, and prescriptions
4. Medical Records
    - Maintain a complete, chronological medical history of each patient
    - Store test results, prescriptions, imaging data, and treatment plans securely
    - Access records for review by authorized medical personnel only
5. Inpatient Management
    - Track patient admissions, assigned beds, ward details, and transfer history
    - Update progress notes, surgery details, and nursing care reports
6. Billing and Payments
    - Generate automated invoices for consultations, treatments, tests, and medicines
    - Accept various modes of payment (cash, card, insurance)
    - Generate discharge bills and clearance certificates